

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE CODE: ITA 0611

COURSE NAME: MACHINE LEARNING

COURSE OUTCOME COVERED

CO4: K- Nearest Neighbour Learning – Locally weighted Regression – Radial Bases Functions – Case Based Learning **(BL 5)**

Assessment Tool Weightage: 20%

QUESTIONS: 2

Total Marks:20

Annexure A

Industry Problem Solving

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Short Answer Test

11. A telecommunications company wants to recommend suitable data plans to customers based on their usage patterns. Design a solution using K-Nearest Neighbour or Case-Based Learning. Explain how similarity between users is measured, how recommendations are generated, and how the system adapts to changing user behavior over time.

Answer:

1. Approach:

- Use K-Nearest Neighbour (KNN) or Case-Based Learning (CBL) to identify customers with similar usage patterns.
- Features: monthly data usage (GB), call minutes, SMS count, peak usage time, recharge frequency.

2. Similarity Measurement:

- Apply Euclidean distance or Cosine similarity between feature vectors.
- Example: Two customers with similar data usage and call patterns will have a small distance → high similarity.

3. Recommendation Generation:

- For a new customer, find the K nearest neighbors (similar users).
- Recommend the most popular plan among those neighbors.
- In CBL, retrieve past cases (users + chosen plans) and adapt them to the new user's profile.

4. Adaptation to Changing Behavior:

- Continuously update the dataset with recent usage.
- Recompute similarities periodically.
- Use sliding windows or weighted similarity (recent behavior given more weight).

Conclusion:

KNN/CBL enables personalized plan recommendations by leveraging similarity in usage patterns, ensuring adaptability as customer behavior evolves.

15. A smart city system aims to predict traffic flow and congestion levels using data from sensors, weather, and time of day. Design a solution using Locally Weighted Regression or KNN. Explain how the model captures dynamic patterns, how predictions are made in real-time, and how scalability is maintained with large volumes of data.

Answer:

1. Approach:

- Use Locally Weighted Regression (LWR) or KNN regression to predict traffic flow.
- Features: sensor counts (vehicles/min), weather (rain, temperature), time of day, day of week.

2. Capturing Dynamic Patterns:

- LWR assigns higher weights to nearby data points (recent traffic conditions).
- KNN regression uses neighboring time slots/locations to estimate congestion.
- Both methods adapt to local variations (rush hour vs. off-peak).

3. Real-Time Predictions:

- Sensor data streams are processed continuously.
- For each query (e.g., traffic at 8:30 AM), the model finds similar past conditions and predicts flow.
- LWR ensures smooth predictions by fitting a local regression curve.

4. Scalability:

- Use distributed computing frameworks (e.g., Spark MLlib) to handle large sensor datasets.
- Apply approximate nearest neighbor search for faster KNN.
- Store historical data in scalable databases (NoSQL, time-series DB).

Conclusion:

LWR/KNN provides adaptive, real-time traffic predictions by focusing on local conditions, while scalable architectures ensure performance across millions of sensor inputs in a smart city.